ℂomplex Eigenvalue Field extensions of Fibonacci Matrices

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Abstract

The Fibonacci sequence, $\mathcal{F}_n$, is well-known for its closed-form expression, Binet's formula, which links the sequence to the Golden Ratio, ϕ. This paper explores the algebraic framework of Fibonacci matrices M, focusing on their diagonalization via characteristic polynomials. Extending M into the complex field $\mathbb{M}$ enables the study of eigenvalues, eigenvectors, and Jordan decomposition. The recursive nature of the sequence is shown to arise from the determinant operation on $n \times n$ complex tridiagonal matrices, with eigenvalues reflecting the structure of algebraic field extensions of $\mathbb{Q}$. For prime n, the eigenvalues suggest a hierarchy of radical expressions, while for composite n, the structure invites further exploration through Galois symmetries and computational approaches. This primarily pedagogical piece introduces standard algebraic operations in novel contexts and provides an entry point for further study of field extensions and polynomial properties.

The **Fibonacci sequence**, $\mathcal{F}_n$ possesses a closed-form expression known as **Binet's formula**, which represents each Fibonacci number as a function of the Golden Ratio $\phi = \frac{1+\sqrt{5}}{2}$, encapsulating the sequence's asymptotic behavior towards ϕ. Fibonacci matrices, M provide an algebraic framework to analyze and compute Fibonacci numbers efficiently. This paper focuses on the diagonalisation of M, through its **characteristic polynomial**. An extension into complex fields, $\mathbb{M}$ affords an exploration of eigenvalues, eigenvectors, and the process of Jordan decomposition.

$\mathcal{F}_n$'s second cousin, the **Pell sequence**, $\mathcal{S}_n$ is useful in various number-theoretic contexts, including the approximation of square roots and the construction of Pell primes, which satisfy certain divisibility properties that make them of interest in primality testing [1]. $\mathcal{S}_n$ also has a closed-form expression similar to Binet's formula,[2].

Both sequences provide opportunities to inform the inquisitive high school thinker of some standard operations applied in novel circumstances. As such this is mostly a pedagogical piece with a few suggestive elements from **algebraic field extension** theory that invite the reader to explore further. Recursive sequences are generated by the action of the determinant operation on $n \times n$ ℂomplex **tridiagonal matrices** [3]. The **Jordan diagonalised** representation of these matrices is determined by setting their characteristic polynomial to a Fibonacci number, $\mathcal{F}_{n+1}$. The structured complex radical **metallic numbered** form of these $\mathcal{F}_{n+1}$, as the product of the eigenvalues of $\mathbb{M}$ is suggestive, at least for prime n that the roots of these polynomials can be understood by considering them in a hierarchy of algebraic extensions of $\mathbb{Q}$. The eigenvalues of the characteristic polynomials of Fibonacci matrices thus provide an entry point into complex field extensions revealing the integer primes, p as moderately radicalised split rational field extensions. The product of the prime split irreducibles being the scaled p-dimensional volume element of the space transformed by the Fibonacci matrix. While those with compound n though not simplified with a python algorithm, generalEigenVectorFibonacci.ipynb may also prove amenable to a Galois type symmetry investigation of their polynomial (field extended) properties.

Characteristic Matrices for Metallic Sequences

The Fibonacci sequence $\mathcal{F}_n$ defined by the recurrence relation, $\mathcal{F}_n = \mathcal{F}_{n-1} + \mathcal{F}_{n-2}$, with initial conditions $\mathcal{F}_0 = 0$ and $\mathcal{F}_1 = 1$ can be generated with the repeated application of the golden **Fibonacci matrix** M_1 :

$$M_1 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \tag{1}$$

on the vector $\begin{pmatrix} \mathcal{F}_{n+1} \\ \mathcal{F}_n \end{pmatrix}$ as:

$$\begin{pmatrix} \mathcal{F}_{n+1} \\ \mathcal{F}_n \end{pmatrix} = M_1^n \begin{pmatrix} \mathcal{F}_1 \\ \mathcal{F}_0 \end{pmatrix}.$$

The **characteristic polynomial** of M_1:

$$\det(M_1 - \lambda 1) = \begin{vmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 - \lambda - 1,$$

solves the *characteristic equation*, $\lambda^2 - \lambda - 1 = 0$, with **eigen value** roots:

$$\lambda_1 = \frac{1+\sqrt{5}}{2}, \quad \lambda_2 = \frac{1-\sqrt{5}}{2}.$$

The Fibonacci generator matrix (1) can be diagonalized as:

$$M_1 = PDP^{-1},$$

where P is the matrix of eigenvectors, and D is the diagonal matrix of eigenvalues. See appendix for details. Complex Fibonacci matrices introduce imaginary components in the recurrence relations:

$$\mathbb{M} = \begin{bmatrix} 1 & i \\ i & 1, \end{bmatrix}$$

with eigenvalues: $\lambda = 1 \pm i$. from a characteristic polynomial:

$$\det(\mathbb{M} - \lambda I) = \lambda^2 - 2\lambda + 2,$$

We can generalize the Fibonacci recurrence using a *metallic* multiplier parameter m that adjusts the contribution of the prior term in the sequence. The modified recurrence relation takes the form:

$$\mathcal{S}_n = m \cdot \mathcal{S}_{n-1} + \mathcal{S}_{n-2},$$

or through a matrix M_m that varies depending on the choice of m:

$$M_m = \begin{pmatrix} m & 1 \\ 1 & 0 \end{pmatrix}.$$

With $m = 2$ and initial values $S_0 = 1$ and $S_1 = 1$, the matrix M_2 produces the Pell (silver) sequence of numbers:

$$\begin{pmatrix} \mathcal{S}_2 \\ \mathcal{S}_1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} \mathcal{S}_3 \\ \mathcal{S}_2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ 3 \end{pmatrix}$$

The resulting sequence for $\mathcal{S}_n$ with $m = 2$ yields the first five terms of the Pell sequence: $\mathcal{S}_0 = 1$, $\mathcal{S}_1 = 1$, $\mathcal{S}_2 = 3$, $\mathcal{S}_3 = 7$, $\mathcal{S}_4 = 17$, and $\mathcal{S}_5 = 41$. The solution of the characteristic equation arising from the characteristic polynomial of M_2:

$$\det(M_2 - \lambda I) = \begin{vmatrix} 2-\lambda & 1 \\ 1 & -\lambda \end{vmatrix},$$

(where I is the identity matrix), has roots the silver ratio of the Pell sequence:

$$\lambda_1 = 1 + \sqrt{2}, \quad \lambda_2 = 1 - \sqrt{2}.$$

Complex Matrix Generator of the Fibonacci Sequence

Geometrically the **determinant** of a matrix of row vectors, $\mathbf{a}, \mathbf{b}, \mathbf{c}$ expresses their their cross and dot products, which for $3d$ looks like:

$$\det \begin{pmatrix} \mathbf{a} \\ \mathbf{b} \\ \mathbf{c} \end{pmatrix} = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}),$$

where the *cross product* $\mathbf{b} \times \mathbf{c}$ produces a vector orthogonal to both $\mathbf{b}$ and $\mathbf{c}$, and the *dot product* $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ measures how much of $\mathbf{a}$ aligns with this orthogonal vector. Consider first the simplest case of a 2 row, $\mathbb{C}$omplex-valued square matrix $\mathbb{M}(2)_m$:

$$\mathbb{M}(2)_m = \begin{pmatrix} 1 & m \\ m & 1 \end{pmatrix}, \tag{2}$$

whose determinant is, $\det(\mathbb{M}(2)_m = 1 \times 1 - m \times m = 1 - m^2$. We require this to yield the third Fibonacci number $F_3 = 2$, so: $1 - m^2 = 2 \Rightarrow m = i$, and we have as our 2×2 Fibonacci matrix:

$$\mathbb{M}(2)_1 = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}. \tag{3}$$

The off-diagonal elements i represent a phase shift, analogous to the anti-symmetric interactions in the cross-product. We extend this to a 3×3 matrix $\mathbb{M}(3)_1 \equiv \mathbb{M}_3$ to ensure that the matrix begins to propagate the Fibonacci sequence correctly:

$$\mathbb{M}(3)_1 = \begin{pmatrix} 1 & m & 0 \\ m & 1 & m \\ 0 & m & 1 \end{pmatrix}.$$

Setting the determinant of $\mathbb{M}_3$ to 3 (the fourth Fibonacci number) we have,

$$\det(\mathbb{M}(3)_1 =) = (1 - m^2) - m(m - 0) = 3, \quad \Rightarrow m = \pm i.$$

Extension to higher *metallic numbers*, m take the form of the metallic matrix $\mathbb{M}(n)_m)$, for m the first term lag multiplier of the metallic series is:

$$\mathbb{M}_n(m) = \begin{pmatrix} 1 & mi & 0 & \cdots & 0 \\ mi & 1 & mi & \cdots & 0 \\ 0 & mi & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & mi \\ 0 & 0 & 0 & mi & 1 \end{pmatrix}.$$

For the Pell sequence (silver series, $m = 2$) the $n = 3$ matrix reads:

$$\mathbb{M}(3)_2 = \begin{pmatrix} 1 & 2i & 0 \\ 2i & 1 & 2i \\ 0 & 2i & 1 \end{pmatrix},$$

and whose determinant yields the fourth number of the Pell sequence proper:

$$\det(\mathbb{M}(3)_2) = 1(1 \times 1 - 2i \times 2i) - 2i(2i \times 1 - 2i \times 0) = 5.$$

Complex Binet & diagonalized Jordan form

To diagonalize the matrix (3), $\mathbb{M}(2)_1 = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$, we start by finding the eigenvalues $\lambda = 1 \pm i \in \mathbb{Q}(i)$ (that are solutions, to the characteristic polynomial):

$$\det(\mathbb{M}(2)_1 - \lambda I) = \det \begin{pmatrix} 1 - \lambda & i \\ i & 1 - \lambda \end{pmatrix} = (1 - \lambda)^2 - i^2 = \lambda^2 - 2\lambda + 2.$$

To find the eigenvector corresponding to $\lambda_1 = 1 + i$, we solve $(\mathbb{M}(2)_1 - \lambda_1 I)\mathbf{v} = 0$.

$$\mathbb{M}(2)_1 - \lambda_1 I = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} - \begin{pmatrix} 1+i & 0 \\ 0 & 1+i \end{pmatrix} = \begin{pmatrix} -i & i \\ i & -i \end{pmatrix}.$$

Let $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$, then:

$$\begin{pmatrix} -i & i \\ i & -i \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This gives the equations:

$$-iv_1 + iv_2 = 0 \quad \Rightarrow \quad v_1 = v_2.$$

Thus, an eigenvector for $\lambda_1 = 1 + i$ is $\mathbf{v}_{\lambda_1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Similarly, $\mathbf{v}_{\lambda_2} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

Since $\mathbb{M}(2)_1$ has distinct eigenvalues, we look to express it as $\mathbb{M}(2)_1 = PJP^{-1}$ being diagonalizable[1] as J, by defining P from the column eigen vectors and J_2 from its two eigenvalues:

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 1+i & 0 \\ 0 & 1-i \end{pmatrix}.$$

The determinant of J_2 is the product of its diagonal entries:

$$\det = (1-i)\cdot(1+i) = 1-(i)^2 = 1+1 = 2.$$

We determine P^{-1} given the determinant, $\det(P) = 1\cdot(-1) - 1\cdot 1 = -2$ as

$$P^{-1} = \frac{1}{\det(P)} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Expanding PDP^{-1}:

$$\mathbb{M}(2)_1 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1+i & 0 \\ 0 & 1-i \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}.$$

Field Splitting within $\det \mathbb{M}_n$

Now starting from the complex matrix generator for Fibonacci (we will drop the subscript 1 in the following as all sequence are Fibonacci-metal, m=1):

$$\mathbb{M}(3)_1 \equiv \mathbb{M}_3 = \begin{pmatrix} 1 & i & 0 \\ i & 1 & i \\ 0 & i & 1 \end{pmatrix},$$

we now seek its Jordan form J_3, as well as the transformation matrix P such that $\mathbb{M}_3 = PJ_3P^{-1}$. The characteristic polynomial for $\mathbb{M}_3$ reads:

$$\lambda^3 - 3\lambda^2 + 5\lambda - 3 = 0$$

Solving this polynomial yields one real and two complex **silver ratio** eigenvalues:

$$\lambda_1 = 1, \quad \lambda_2 = 1 - \sqrt{2}\cdot i, \quad \lambda_3 = 1 + \sqrt{2}\cdot i$$

Since each eigenvalue has multiplicity 1, the Jordan form J_3 is a diagonal matrix:

$$J_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1-\sqrt{2}\cdot i & 0 \\ 0 & 0 & 1+\sqrt{2}\cdot i \end{pmatrix}.$$

The determinant of J_3, $\in \mathbb{Q}(i, \sqrt{2})$ is the product of its diagonal entries:

$$\det J_3 = 1\cdot\left(1-\sqrt{2}\,i\right)\cdot\left(1+\sqrt{2}\,i\right) = 3.$$

.
The eigenvectors corresponding to each eigenvalue are:

$$\mathbf{v}_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix} \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix},$$

forming the columns of the matrix P which has an inverse P^{-1}:

$$P = \begin{pmatrix} -1 & 1 & 1 \\ 0 & -\sqrt{2} & \sqrt{2} \\ 1 & 1 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} -\frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{4} & -\frac{\sqrt{2}}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{\sqrt{2}}{4} & \frac{1}{4} \end{pmatrix}$$

[1]For higher order or complex matrices, Jordan decomposition generalizes the diagonalization, $\mathbb{M} = PJP^{-1}$.

To confirm the Jordan decomposition, we can calculate PJP^{-1} and verify that it reconstructs $\mathbb{M}_3$. You can do this and find subsequent diagonal matrices by using the python code, generalEigenVectorFibonacci.ipynb.

$$\det(J_n) = \mathcal{F}_{n+1}.$$

We can proceed with this analysis gradually building up the the order, n of the characteristic nth order polynomial as shown in Table 1 and noting that the structure of the prime number Matrices, $\mathbb{M}_p$ appear to be simpler than compound $\mathbb{M}_n$.

n	**Characteristic Polynomial**	$\mathcal{F}_{n+1}$
2	$\lambda^2 - 2\lambda + 2$	2
3	$\lambda^3 - 3\lambda^2 + 5\lambda - 3$	3
4	$\lambda^4 - 4\lambda^3 + 9\lambda^2 - 10\lambda + 5$	5
5	$\lambda^5 - 5\lambda^4 + 14\lambda^3 - 22\lambda^2 + 15\lambda - 5$	8
6	$\lambda^6 - 6\lambda^5 + 20\lambda^4 - 40\lambda^3 + 42\lambda^2 - 24\lambda + 6$	13
7	$\lambda^7 - 7\lambda^6 + 27\lambda^5 - 65\lambda^4 + 99\lambda^3 - 84\lambda^2 + 36\lambda - 7$	21
8	$\lambda^8 - 8\lambda^7 + 35\lambda^6 - 98\lambda^5 + 175\lambda^4 - 180\lambda^3 + 105\lambda^2 - 28\lambda + 8$	34
9	$\lambda^9 - 9\lambda^8 + 44\lambda^7 - 140\lambda^6 + 285\lambda^5 - 378\lambda^4 + 330\lambda^3 - 180\lambda^2 + 45\lambda - 9$	55
10	$\lambda^{10} - 10\lambda^9 + 54\lambda^8 - 192\lambda^7 + 440\lambda^6 - 704\lambda^5 + 792\lambda^4 - 600\lambda^3 + 300\lambda^2 - 90\lambda + 10$	89
11	$\lambda^{11} - 11\lambda^{10} + 65\lambda^9 - 250\lambda^8 + 660\lambda^7 - 1210\lambda^6 + 1650\lambda^5 - 1650\lambda^4 + 1210\lambda^3 - 660\lambda^2 + \cdots$	144

Table 1: Characteristic Polynomials and Fibonacci Numbers

$p \times p$ Matrices

We have that the 5×5 matrix includes complex **bronze ratio**-like eigenvalues, $\lambda \in \mathbb{Q}(i, \sqrt{3})$:

$$J_5 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1-i & 0 & 0 & 0 \\ 0 & 0 & 1+i & 0 & 0 \\ 0 & 0 & 0 & 1-\sqrt{3}\,i & 0 \\ 0 & 0 & 0 & 0 & 1+\sqrt{3}\,i \end{pmatrix}$$

The determinant of J_5, $\det J_5 = (1) \cdot (1-i) \cdot (1+i) \cdot \left(1-\sqrt{3}\,i\right) \cdot \left(1+\sqrt{3}\,i\right) = 8$.
The 7×7 matrix includes **silver ratio**, $\lambda \in \mathbb{Q}(i, \sqrt{2})$ from J_3 and $\mp\sqrt{-2 \mp \sqrt{2}} = \pm i\sqrt{2 \mp \sqrt{2}}$:

$$J_7 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1-\sqrt{2}\,i & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1+\sqrt{2}\,i & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1-i\sqrt{2+\sqrt{2}} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1+i\sqrt{2+\sqrt{2}} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1-i\sqrt{2-\sqrt{2}} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1+i\sqrt{2-\sqrt{2}} \end{pmatrix}$$

Whereas for $n = 4, 8$ we need to define the imaginary irrational coefficients

$$\alpha_\pm = \frac{i}{\sqrt{2}}\sqrt{3 \pm \sqrt{5}}, \quad \alpha_\pm^* = -\frac{i}{\sqrt{2}}\sqrt{3 \pm \sqrt{5}} \tag{4}$$

$$\beta_\pm = \frac{i}{\sqrt{2}}\sqrt{5 \pm \sqrt{5}}, \quad \beta_\pm^* = -\frac{i}{\sqrt{2}}\sqrt{5 \pm \sqrt{5}} \tag{5}$$

so that

$$J_4 = \begin{pmatrix} 1 - \frac{\sqrt{2}\,i\sqrt{3-\sqrt{5}}}{2} & 0 & 0 & 0 \\ 0 & 1 + \frac{\sqrt{2}\,i\sqrt{3-\sqrt{5}}}{2} & 0 & 0 \\ 0 & 0 & 1 - \frac{\sqrt{2}\,i\sqrt{\sqrt{5}+3}}{2} & 0 \\ 0 & 0 & 0 & 1 + \frac{\sqrt{2}\,i\sqrt{\sqrt{5}+3}}{2}. \end{pmatrix}$$

with $\det J_4 = \left(1 + \frac{3-\sqrt{5}}{2}\right) \cdot \left(1 + \frac{\sqrt{5}+3}{2}\right) = 5$ can be written as

$$J_4 = \begin{pmatrix} 1+i\alpha_+ & 0 & 0 & 0 \\ 0 & 1-i\alpha_+ & 0 & 0 \\ 0 & 0 & 1+i\alpha_- & 0 \\ 0 & 0 & 0 & 1-i\alpha_- \end{pmatrix},$$

while J_8 with determinant 34 reads

$$J_8 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \\ 0 & 1+i\beta_+ & 0 & 0 & 0 & 0 & 0 & 0 & \\ 0 & 0 & 1-i\beta_+ & 0 & 0 & 0 & 0 & 0 & \\ 0 & 0 & 0 & 1+i\beta_- & 0 & 0 & 0 & 0 & \\ 0 & 0 & 0 & 0 & 1-i\beta_- & 0 & 0 & 0 & \\ 0 & 0 & 0 & 0 & 0 & 1-i\beta_+ & 0 & 0 & \\ 0 & 0 & 0 & 0 & 0 & 0 & 1+i\beta_+ & 0 & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1-i\beta_- & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1+i\beta_- \end{pmatrix}.$$

Clarke [5] explores the simplification of square roots of the form $\sqrt{a \pm b\sqrt{m}}$, outlining the connection between Pell's equation and the square roots of surds given the simplification condition:

$$a^2 - mb^2 = n^2, \tag{6}$$

where a, b, and m are integers, m is square-free, and n is an integer. If this condition holds, the surd $\sqrt{a \pm b\sqrt{m}}$ can be expressed as:

$$\sqrt{a \pm b\sqrt{m}} = \frac{\sqrt{a+n} \pm \sqrt{a-n}}{2}.$$

We apply this framework to examine whether simplifications exist for 5.

Case: $\sqrt{3 \pm \sqrt{5}}$ $:a = 3$, $b = 1$, $m = 5$ so $a^2 - mb^2 = 3^2 - 5 \cdot 1^2 = 9 - 5 = 4 = 2^2$. so since $n = 2$:

$$\sqrt{3 \pm \sqrt{5}} = \frac{\sqrt{3+2} \pm \sqrt{3-2}}{2} = \frac{\sqrt{5} \pm 1}{2},$$

which are the golden ratio $\phi = \frac{1+\sqrt{5}}{2}$, and its conjugate $\phi' = \frac{1-\sqrt{5}}{2}$ so: $\alpha_+ = \frac{i}{\sqrt{2}}\phi, \quad \alpha_- = \frac{i}{\sqrt{2}}\phi'$, and we have:

$$J_4 = \begin{pmatrix} 1 + \frac{i}{\sqrt{2}}\phi & 0 & 0 & 0 \\ 0 & 1 - \frac{i}{\sqrt{2}}\phi & 0 & 0 \\ 0 & 0 & 1 + \frac{i}{\sqrt{2}}\phi' & 0 \\ 0 & 0 & 0 & 1 - \frac{i}{\sqrt{2}}\phi' \end{pmatrix},$$

The condition for 6 to always be a perfect square n^2 is that a, b satisfy Pell's equation:

$$x^2 - my^2 = 1,$$

where $x = a$ and $y = b$. Note that if $a = b$, then we require, $a^2(1 - m) = n^2.m = 1$ is the only case where $a = b$ satisfies the condition with $n^2 = 0$, which is trivial and thus $\sqrt{5 \pm \sqrt{5}}$ within β cannot be simplified.

The code loopCharcteristic1.ipynb delivers the Table 2 of eigenvalues of the characteristic equation of the complex Fibonacci matrix, where n corresponds to the rank of the matrix. Geometrically recall that these eigenvalues, λ represent the scalar factors by which the matrix (merely) scales its eigenvectors under transformation. Each eigenvalue corresponds to a stretching (or compressing) factor in the direction of its eigenvector. For the original matrix $\mathbb{M}$, the action of the matrix on an eigenvector v satisfies $\mathbb{M}v = \lambda v$, where λ is the eigenvalue. . The determinant of D_n as the product of eigenvalues represents the scaled n-dimensional volume element of the space

n	**Eigenvalues**, λ	$\mathbf{Fib}_{n+1}$
2	$[1-i, 1+i]$	2
3	$[1, 1-\sqrt{2}i, 1+\sqrt{2}i]$	3
4	$\left[1 \mp \frac{i}{\sqrt{2}}\phi \text{ or } \phi'\right]$	5
5	$[1, 1-i, 1+i, 1-\sqrt{3}i, 1+\sqrt{3}i]$	8
6	$\left[1 - \sqrt{\frac{-60-6\cdot 28^{1/3}(1+3\sqrt{3}i)}{2}}, \dots\right]$	13
7	$[1, 1-\sqrt{2}i, 1+\sqrt{2}i, 1 \pm i\sqrt{2 \mp \sqrt{2}}$	21
8	$\left[1-i, 1+i, \text{CRootOf}(x^6 - 6x^5 + 21x^4 - \dots)\right]$	34
9	$\left[1, 1 \mp \frac{i\sqrt{10 \pm 2\sqrt{5}}}{2}, \dots\right]$	55
10	$\left[1 - \sqrt{10}\sqrt{-72 - 2\cdot 11^{1/5}\cdot 2^{1/10}}, \dots\right]$	89
11	$\left[1, 1-i, 1+i, 1-\sqrt{2}i, 1+\sqrt{2}i, \dots\right]$	144

Table 2: Eigenvalues and Fibonacci Numbers

transformed by the Fibonacci matrix:

$$n = 2 : \prod_{\lambda_2} = (1-i)(1+i) = 2, \tag{7}$$

$$n = 3 : \prod_{\lambda_3} = 1 \cdot (1-\sqrt{2}i)(1+\sqrt{2}i) = 3, \tag{8}$$

$$n = 4 : \prod_{\lambda_4} = \left[1 - \frac{i}{\sqrt{2}}\phi\right]\left[1 + \frac{i}{\sqrt{2}}\phi\right]\left[1 - \frac{i}{\sqrt{2}}\phi'\right]\left[1 + \frac{i}{\sqrt{2}}\phi'\right] = 5,$$

$$n = 5 : \prod_{\lambda_5} = 1 \cdot [1-i]\,[1+i]\,2\Phi' \cdot 2\Phi = 4\prod_{\lambda_2} \Phi' \cdot \Phi = 8.$$

$$n = 7 : \prod_{\lambda_6} = 1 \cdot (1-\sqrt{2}i)(1+\sqrt{2}i)(1 - i\sqrt{2-\sqrt{2}})(1 + i\sqrt{2-\sqrt{2}})\cdots,$$

$$= \prod_{\lambda_3} (1 - i\sqrt{2-\sqrt{2}})(1 + i\sqrt{2-\sqrt{2}})\cdots = 21 \tag{9}$$

$$n = 11 : \prod_{\lambda_7} = 1 \cdot (1-i)(1+i)(1-\sqrt{2}i)(1+\sqrt{2}i) = \prod_{\lambda_2} \cdot \prod_{\lambda_3} \cdots = 144. \tag{10}$$

for complex golden ratio (Φ) arising from the quadratic equation $x^2 - x + 1 = 0$. :

$$\Phi = \frac{1 + i\sqrt{3}}{2}, \quad \Phi' = \frac{1 - i\sqrt{3}}{2}, \tag{11}$$

For prime $n \in \mathbb{N}$, these can be viewed as $\mathbb{C}$omplex **field extension** decompositions. The primes as eigenvalues are thus intertwined with polynomial characteristics through the action of the determinant on the Fibonacci-Jordan matrix. This is to be expected as the breaking down of polynomials into irreducible factors parallels decomposing composite numbers into prime factors.

Accordingly we gather some observations of the Galois theory of field extensions:

1. **Decomposition of Composite Numbers and Polynomials:**
   Polynomials $f(x) \in F[x]$ over a **field** F can be expressed as a product of irreducible polynomials, as integers can be expressed as a product of primes. While composite numbers n decompose uniquely into prime factors:

   $$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}.$$

   composite polynomials $f(x)$ decompose uniquely (up to units) into irreducible factors:

   $$f(x) = p_1(x)^{e_1} p_2(x)^{e_2} \cdots p_k(x)^{e_k}.$$

2. **Factorization and Rings:** analogous to the **fundamental theorem of arithmetic**, the **unique factorization theorem** ensures this decomposition in $F[x]$. For integers, factorization occurs in the **ring** $\mathbb{Z}$, where primes are indivisible except by ± 1 and themselves. Extending to the **Gaussian integers** $\mathbb{Z}[i]$, which include complex numbers of the form $a + bi$ (where $a, b \in \mathbb{Z}$ and $i = \sqrt{-1}$), the notion of primes changes.

3. **Root extensions and splits:** integers $\mathbb{Z}$-primes either split or remain inert within **field extensions**, depending on the properties of the extension. In $\mathbb{Z}[i]$, $\mathbb{Z}$-primes are redefined to account for the new units $\pm i$, so that factorization by **Gaussian primes** means 2, 5 lose their primality status:

$$2 = (1 + i)(1 - i), \quad 5 = (2 + i)(2 - i).$$

   Primes in $\mathbb{Z}[i]$ are thus classified into:

   (a) **Real primes** that remain irreducible (e.g., 3 retains its primality, remaining inert in $\mathbb{Z}[i]$ as it cannot be decmposed into Gaussian prime factors.

   (b) **Decomposable primes**, such as 2 and 5, which split into products of Gaussian primes.

   Extending further to $\mathbb{Q}(i, \sqrt{2})$, we lose the primality of 3. The eigenvalues $[1, 1 - \sqrt{2}i, 1 + \sqrt{2}i]$ for $n = 3$ involve $\sqrt{2}$ and i so 3 is factorizable in $\mathbb{Q}(i, \sqrt{2})$ (in accordance with the decomposition of its characteristic polynomial into linear terms).

   For polynomials, factorization occurs in $F[\lambda]$, but the irreducibility depends on the field F. For instance, $\lambda^2 - 2$ is irreducible over $\mathbb{Q}$, the field of rational numbers but factors as $(\lambda - \sqrt{2})(\lambda + \sqrt{2})$ over $\mathbb{R}$:

$$\lambda^2 - 2 \quad \text{(irreducible over } \mathbb{Q}\text{) splits as} \quad (\lambda - \sqrt{2})(\lambda + \sqrt{2}) \quad \text{over } \mathbb{R}.$$

4. **Galois Theory:** formalizes these ideas by studying the symmetry of roots:

$$M \sim \text{Primes} \quad \text{and} \quad \text{Galois Group} \sim \text{Field Extensions}.$$

   A composite number's prime factorization reflects its divisors and the structure of its multiplicative group. Similarly, the splitting field of a polynomial reflects how its roots relate under symmetries. The **Galois group** of a polynomial captures the structure of the field extension required for its roots, akin to how composite numbers decompose into primes.

Eisenstein Integers

The Eisenstein integers, denoted $\mathbb{Z}[\omega]$, form a quadratic integer ring defined as:

$$\mathbb{Z}[\omega] = \{r + s\omega : r, s \in \mathbb{Z}\},$$

where $\omega = e^{2\pi i/3}$ is a primitive third root of unity. The root ω satisfies:

$$\omega^3 = 1, \quad \omega^2 + \omega + 1 = 0, \quad \omega^2 = -1 - \omega.$$

The Eisenstein integers are a natural generalization of the Gaussian integers $\mathbb{Z}[i]$, which are formed by adjoining i (a fourth root of unity) to $\mathbb{Z}$. Similarly, the Eisenstein integers are formed by adjoining ω (a third root of unity) to $\mathbb{Z}$. The roots $\Phi = \omega$ and $\Phi' = \omega^2$ of the quadratic equation: $x^2 - x + 1 = 0$, divide the unit circle into three equal arcs and geometrically form an equilateral triangle with vertices at $1, \omega, \omega^2$. To ensure closure under division, Eisenstein integers extend the ring $\mathbb{Z}[\omega]$ to the field $\mathbb{Q}(\omega, \sqrt{3})$, defined as:

$$\mathbb{Q}(\omega, \sqrt{3}) = \left\{ \frac{r + s\omega}{2} : r, s \in \mathbb{Z} \right\}.$$

This field allows division by 2, which is necessary for accommodating elements such as $\Phi = \frac{1+i\sqrt{3}}{2}$. The Eisenstein integers inherit many properties from the Gaussian integers, with modifications due to the third root of unity ω. For $z = r + s\omega$, where $r, s \in \mathbb{Z}$, the following operations are defined:

- **Addition:** Simply add the components:

$$z_1 + z_2 = (r_1 + r_2) + (s_1 + s_2)\omega.$$

- **Multiplication:** Incorporates the property $\omega^2 = -1 - \omega$:

$$z_1 \cdot z_2 = (r_1 r_2 + s_1 s_2(-1-\omega)) + (r_1 s_2 + r_2 s_1)\omega.$$

- **Norm:** The norm of $z = r + s\omega$ is defined as:

$$N(z) = z \cdot \overline{z} = r^2 - rs + s^2,$$

where $\overline{z} = r - s - s\omega$ is the conjugate of z.

The Eisenstein integers $\mathbb{Z}[\omega]$ generalize the Gaussian integers by introducing the third root of unity ω and its associated algebraic structure. This ring is further extended to $\mathbb{Q}(\omega, \sqrt{3})$ to accommodate closure under division, allowing for deeper exploration of number-theoretic properties. The norm $N(z) = r^2 - rs + s^2$ plays a central role in understanding the structure and arithmetic of Eisenstein integers. The norm $N(z)$ is always a non-negative integer, providing a measure of the "size" of z in $\mathbb{Z}[\omega]$. The property that $N(z)$ is multiplicative ensures that $\mathbb{Z}[\omega]$ is also a unique factorization domain (UFD), just like $\mathbb{Z}[i]$.

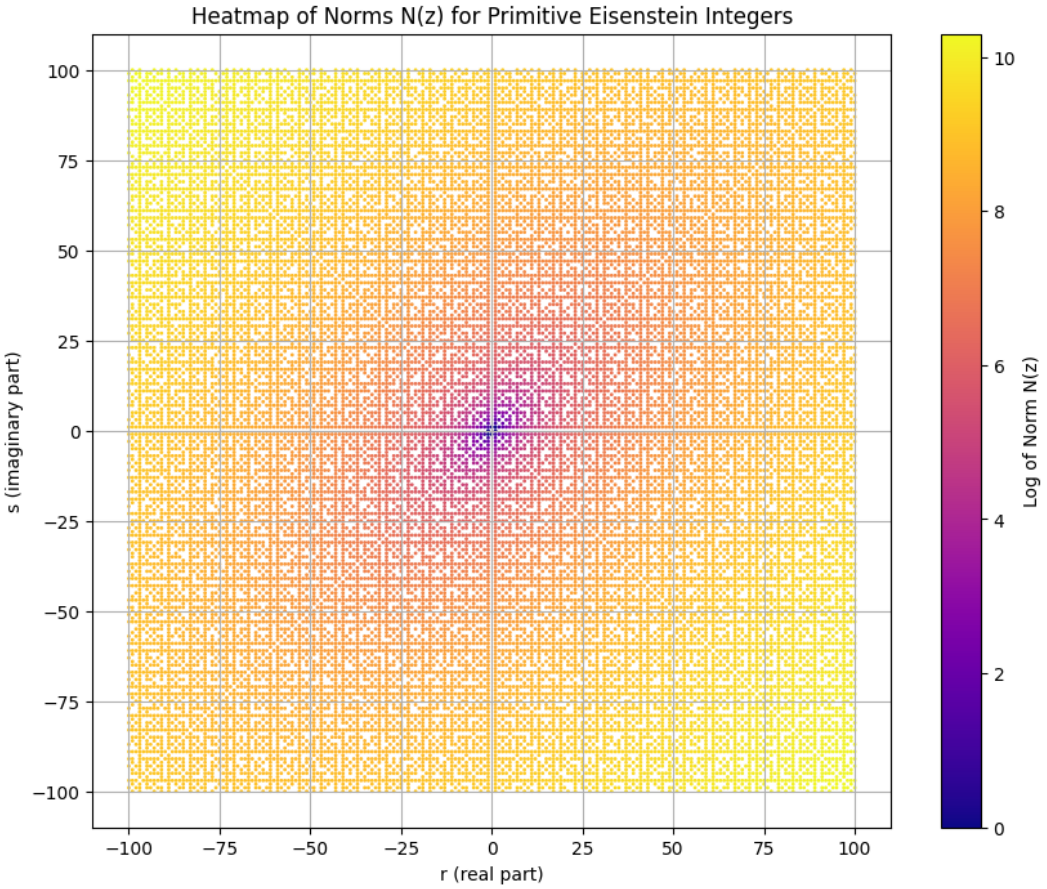


Figure 1: Pythagorean Triples from the elevenths.

The plot visualizes the distribution of **primitive Eisenstein integers** $z = r + s\omega$, where ω is a primitive third root of unity ($\omega^3 = 1$). The key elements of the plot are:

- Each point (r, s) represents the real part r and imaginary coefficient s of Eisenstein integer z.
- The Eisenstein integers shown are **primitive**, meaning they satisfy the condition:

$$\gcd(a, b, c) = 1,$$

where $a = r$, $b = -r + s$, and $c = r^2 - rs + s^2$ (the norm of z).

- The **color intensity** of each point corresponds to the logarithm of the norm:

$$N(z) = r^2 - rs + s^2.$$

This scaling highlights variations in the norm across a wide range of values.

Characteristic Polynomial of $\mathbb{M}_2(1)$ with ω

The matrix $\mathbb{M}_2(1)$ is given by:

$$\mathbb{M}_2(1) = \begin{pmatrix} 1 & \omega \\ \omega & 1 \end{pmatrix}.$$

The characteristic polynomial is computed as:

$$\det(\mathbb{M}_2(1) - \lambda I) = \det \begin{pmatrix} 1-\lambda & \omega \\ \omega & 1-\lambda \end{pmatrix}.$$

Expanding the determinant, we have:

$$\det(\mathbb{M}_2(1) - \lambda I) = (1-\lambda)^2 - \omega^2.$$

Using $\omega^2 = -1 - \omega$ (from $\omega^2 + \omega + 1 = 0$), substitute ω^2 into the determinant:

$$(1-\lambda)^2 - (-1-\omega) = (1-\lambda)^2 + 1 + \omega.$$

Simplify:

$$(1-\lambda)^2 + 1 + \omega = \lambda^2 - 2\lambda + 2 + \omega.$$

Thus, the characteristic polynomial is:

$$\lambda^2 - 2\lambda + 2 + \omega = 0.$$

The eigenvalues are given by solving:

$$\lambda = \frac{2 \pm \sqrt{z}}{2}, \quad \text{where } z = -2 - 2i\sqrt{3}.$$

Factor out -2 from z:

$$z = -2(1 + i\sqrt{3}).$$

Express $1 + i\sqrt{3}$ in polar form:

$$1 + i\sqrt{3} = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) = 2e^{i\pi/3}.$$

Thus:

$$z = -2 \cdot 2e^{i\pi/3} = -4e^{i\pi/3}.$$

Using the polar form, the square root of z is:

$$\sqrt{z} = \sqrt{-4e^{i\pi/3}} = \sqrt{-4} \cdot \sqrt{e^{i\pi/3}}.$$

Compute each factor:

$$\sqrt{-4} = 2i, \quad \sqrt{e^{i\pi/3}} = e^{i\pi/6}.$$

Thus:

$$\sqrt{z} = 2i \cdot e^{i\pi/6} = 2i\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right).$$

Simplify further:

$$\sqrt{z} = 2i\left(\frac{\sqrt{3}}{2} + i\frac{1}{2}\right) = i\sqrt{3} - 1.$$

Substitute $\sqrt{z} = i\sqrt{3} - 1$ into the eigenvalue formula:

$$\lambda = \frac{2 \pm (i\sqrt{3} - 1)}{2}.$$

Compute the two eigenvalues:

$$\lambda_1 = \frac{2 + (i\sqrt{3} - 1)}{2} = \frac{1 + i\sqrt{3}}{2},$$
$$\lambda_2 = \frac{2 - (i\sqrt{3} - 1)}{2} = \frac{1 - i\sqrt{3}}{2}.$$

The eigenvalues are:

$$\lambda_1 = \Phi = \frac{1 + i\sqrt{3}}{2}, \quad \lambda_2 = \Phi' = \frac{1 - i\sqrt{3}}{2},$$

which are the **complex golden ratios**. The matrix $\mathbb{M}_2(1)$ diagonalizes as:

$$\mathbb{M}_2(1) = P \begin{pmatrix} \Phi & 0 \\ 0 & \Phi' \end{pmatrix} P^{-1},$$

where P is the matrix of eigenvectors. The eigenvalues are connected to the rotational symmetry of the Eisenstein integers and the complex golden ratios.

Our nuanced interplay between primes over field extensions and the roots of characteristic polynomials can be summarized as:

1. **Field Extensions for Eigenvalues** of the characteristic polynomials for each n of Fibonacci matrix lie in specific algebraic nested extensions of $\mathbb{Q}$:
   - For $n = 2$, the eigenvalues $1 - i$ and $1 + i$ lie in $\mathbb{Q}(i)$, the smallest extension of $\mathbb{Q}$ containing i, the imaginary unit.
   - For $n = 3$, the eigenvalues $1 - \sqrt{2}i$ and $1 + \sqrt{2}i$ lie in $\mathbb{Q}(i, \sqrt{2})$, an extension formed by adjoining i and $\sqrt{2}$ to $\mathbb{Q}$. Although **Primes** $p \equiv 3 \pmod 4$ are also Irreducible in $\mathbb{Q}[i]$: $3, 7, 11, 19, \ldots$ as reflected in the recursion between 7, 8, 9, and 10.
   - For $n = 7$, eigenvalues such as $1 \pm i\sqrt{2 - \sqrt{2}}$ requires the nested radical and the imaginary unit field extension $\mathbb{Q}(i, \sqrt{2}, \sqrt{2 - \sqrt{2}})$.
2. **Inclusion of Smaller Roots:** From Table 2, we observe a hierarchy of eigenvalues where roots from smaller matrices D_2 and D_3 appear within those of larger ones, D_5, D_7, and D_{11}:
   - The eigenvalues of D_2 (e.g., $1 - i$, $1 + i$) are present in D_5 and D_7.
   - The eigenvalues of D_3 (e.g., $1 - \sqrt{2}i$, $1 + \sqrt{2}i$) appear in D_7.
   - All roots from D_2, D_3, D_5, and D_7 are observed in D_{11}.

   This nesting of roots reflects the structure of field extensions, with each D_n encapsulating extensions required by smaller Fibonacci matrices.

The decomposition into complex roots of the eigenvalues of Fibonacci matrices reveals the hierarchical nature of field extensions and provides a rich framework for pedagogical exploration. The Galois groups of these polynomials encode the symmetries of the roots. Thus the polynomials of Table 1 could be further analyzed for their splitting fields, irreducibility, and symmetry properties. While J_p for primes, $p \in \mathbb{N}$ appear structurally simple, composite n order matrices are riddled with multiple stacked irrationals which are impenetrable to python's sympy `simplify` algorithm within generalEigenVectorFibonacci.ipynb. This presumably stems from the lack of divisibility in the order the matrix and could be worth exploring further, in the first instance (after first normalising them) by examining their roots of unity .

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Appendix: Binet's formula from M_1

Following the exposition of Rochford, [4] we have

$$\text{for } M_1 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \text{ so that } \begin{pmatrix} F_2 \\ F_1 \end{pmatrix} = M_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

and for $n \geq 1$,

$$\begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} F_n \\ F_{n-1} \end{pmatrix}, \quad \begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = M_1^n \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad F_n = \begin{pmatrix} 0 & 1 \end{pmatrix} M_1^n \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

$$\begin{pmatrix} F_3 \\ F_2 \end{pmatrix} = M_1 \begin{pmatrix} F_2 \\ F_1 \end{pmatrix} = M_1^2 \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} F_4 \\ F_3 \end{pmatrix} = M_1 \begin{pmatrix} F_3 \\ F_2 \end{pmatrix} = M_1^3 \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

To diagonalize M_1 is to write it as $M_1 = PDP^{-1}$, where D is a diagonal matrix:

$$M_1^2 = PDP^{-1}PDP^{-1} = PD^2P^{-1}.$$

Similarly, for any natural number n, we have $M_1^n = PD^nP^{-1}$. We first calculate its eigenvalues by finding the roots of the characteristic polynomial, $\det(M_1 - \lambda I)$, as

$$\lambda_1 = \frac{1+\sqrt{5}}{2} \quad \text{and} \quad \lambda_2 = \frac{1-\sqrt{5}}{2}.$$

The matrix D is therefore

$$D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}.$$

We note that $\lambda_1 + \lambda_2 = 1$ and $\lambda_1\lambda_2 = -1$. The columns of P are the eigenvectors corresponding to the eigenvalues that form the diagonal entries of $D$and so we calculate a basis for the null space of $M_1 - \lambda I$. For the eigenvalue λ_1,

$$M_1 - \lambda_1 I = \begin{pmatrix} \lambda_2 & 1 \\ 1 & -\lambda_1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & 1-\lambda_1 \end{pmatrix}.$$

The null space of $M_1 - \lambda_1 I$ is spanned by the vector $\begin{pmatrix} \lambda_1 \\ 1 \end{pmatrix}$. To simplify P^{-1}, we normalize this and the normalized eigenvector corresponding to λ_2 to length 1, yielding:

$$\vec{v}_{\lambda_1} = \frac{1}{\sqrt{5}} \begin{pmatrix} \lambda_1 \\ 1 \end{pmatrix}, \quad \vec{v}_{\lambda_2} = \frac{1}{\sqrt{5}} \begin{pmatrix} \lambda_2 \\ 1 \end{pmatrix}.$$

Thus, the matrix P and its inverse given $P^{-1} = P^\top$ in the diagonalization of M_1 is

$$P = \frac{1}{\sqrt{5}} \begin{pmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{pmatrix} \quad P^{-1} = \frac{1}{\sqrt{5}} \begin{pmatrix} \lambda_1 & 1 \\ \lambda_2 & 1 \end{pmatrix}.$$

Finally, we derive the closed-form expression for F_n:

$$F_n = \begin{pmatrix} 0 & 1 \end{pmatrix} M_1^n \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{5}} \left(\lambda_1^n - \lambda_2^n\right).$$

This completes the derivation of Binet's formula for the Fibonacci sequence:

$$F_n = \frac{1}{\sqrt{5}} \left(\lambda_1^n - \lambda_2^n\right).$$